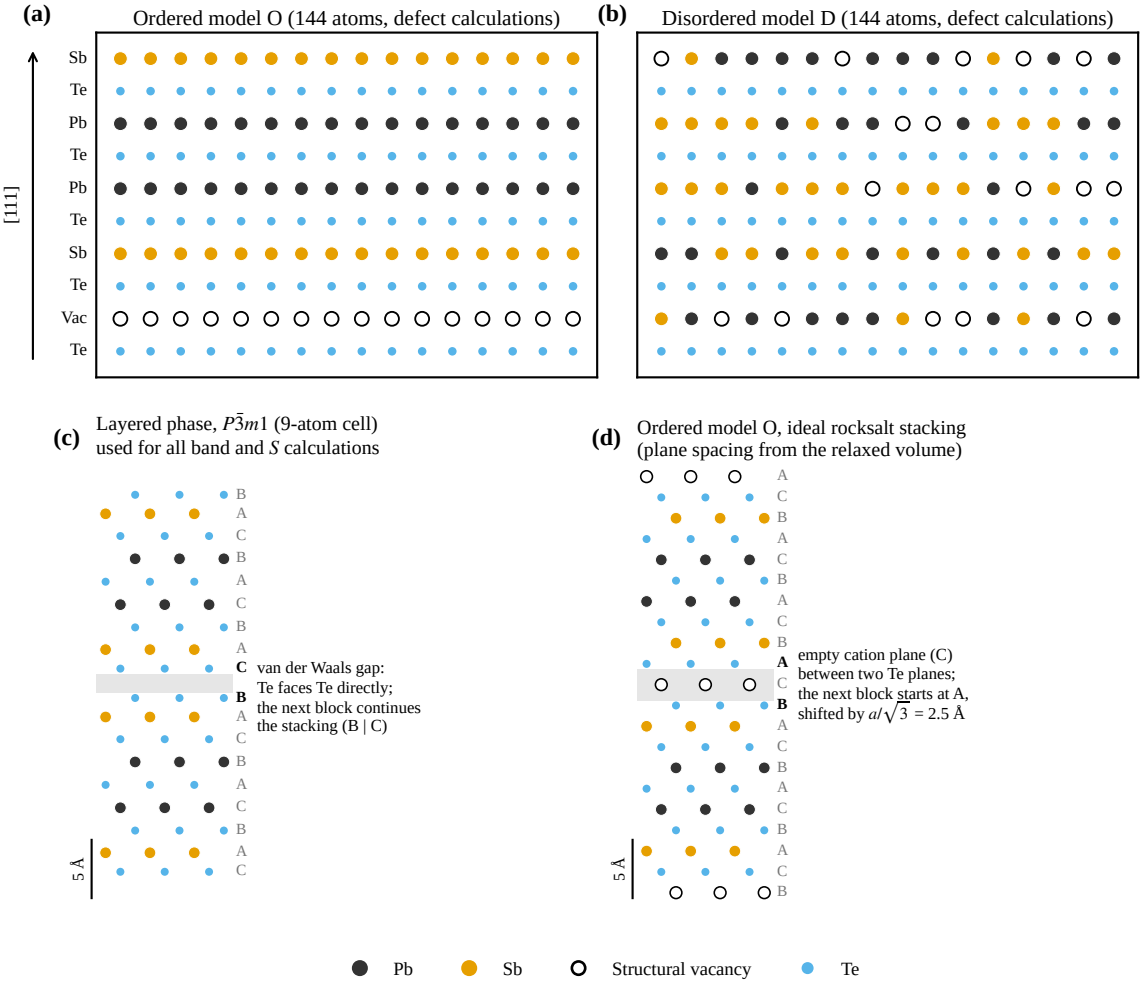

(a)
Ordered model O (144 atoms, defect calculations)
(b)
Disordered model D (144 atoms, defect calculations)
[111]
Sb
Te
Pb
Te
Pb
Te
Sb
Te
Vac
Te
(c)
Layered phase, $P\bar{3}m1$ (9-atom cell)
used for all band and S calculations
B
A
C
B
A
C
B
A
C
B
A
C
van der Waals gap:
Te faces Te directly;
the next block continues
the stacking (B | C)
5 Å
(d)
Ordered model O, ideal rocksalt stacking
(plane spacing from the relaxed volume)
A
C
B
A
C
B
A
C
B
A
C
B
A
C
empty cation plane (C)
between two Te planes;
the next block starts at A,
shifted by $a/\sqrt{3}$ = 2.5 Å
5 Å
Pb
Sb
Structural vacancy
Te